Hacettepe University - Computer Engineering
CMP720 Embedded System Design - Spring 2026

Final Project Topic Proposal Template

(Fill in the sections below. Replace placeholders with your content.)

Group Members (Name Surname + Student ID)

Instructions: List every group member as "Name Surname — Student ID".

- **Yusuf Çağrı Öğüt — N25240126**

Project Topic Title

Instructions: Provide a concise, descriptive title.

[**Hybrid Cellular-LoRa Network Based Agriculture Monitoring and Control Application for Hobby Gardens without Internet Access**]

Project Track (choose one)

■ **Implementation Track**

□ **Optimization Track**

Abstract (300–500 words)

Instructions: In 300–500 words, explain: (i) the problem tackled, (ii) the proposed solution approach and methodology, (iii) high-level system architecture and major components, and (iv) an initial plan and milestones for attacking the problem. Include one crude block diagram (schematic for Implementation track / optimization pipeline diagram for Optimization track) to illustrate the idea.

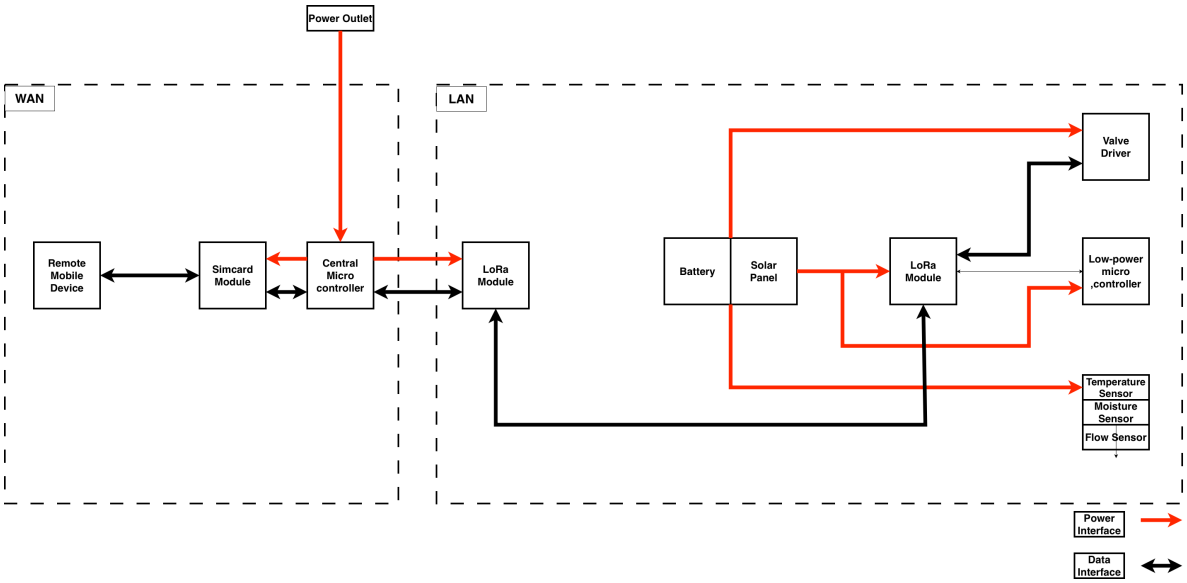
IoT applications for hobby gardens is a trending subject [1], [2], [3]. Many hobby gardens are built in places where internet connection is not available. This creates a problem for implementing IoT applications. Using cellular network is a good workaround for this problem. Commands for valves can be sent and sensor data can be obtained via SMS. For this purpose, a hybrid network which consists of a WAN (Wide Area Network) and LAN (Local Area Network) is proposed. WAN will consist of a remote mobile device to send commands and receive sensory data and a simcard module to be an interface between WAN (remote mobile device) and LAN (central microcontroller).

LAN will consist of a central controller which bridges between simcard module and valves/sensors and their microcontrollers; an antenna for receiving and sending data to/from central controller; a low-power microcontroller; moisture, temperature and flow sensors; battery, solar panel and LoRa modules for short distance communication.

Due to inherent nature of agriculture applications, some modifications and protections should be applied to the hardware. In order to keep the antenna signal at maximum, antenna of LoRa module will be placed 30-40 cm above the ground with a simple plastic pipe [4]. A shielded cable will carry the signal to/from microcontroller. As microcontroller will be located on the ground, buried [5], it will be protected with an appropriate coverage. Digital sensors will be chosen considering the possible voltage drop along the wire (which can be up to 10 meters), so the signal level will be kept.

One of the hardships of the project will be managing power consumption. A preliminary power budget calculation is done. As microcontroller will be in sleep mode mostly, it will work 23.9 hours of day, energy consumption will be 0.14mAh consumptions per day. It is planned that sensors will be checked every 20 minutes. This corresponds to 0.1mAh per day. For energy efficiency a latched-valve will be used, which doesn't require energy for holding it's position. They generally draw 1 amperes in each work cycle and it takes approximately 50 ms. Considering closing and opening motions, it will draw 0.03 mAh per day. LoRa is in deep sleep mode in %99 of times. Therefore, it has a power consumption of 0.7 mAh per day. Total consumption corresponds to around 1 mAh per day. Considering using 10 lines for the system, the total is 30 mAh per day. the battery planned to be is a standard li-po battery with capacity of 3.000 mAh. This means in ideal conditions battery will last 100 days without charging. Note that this might be reduced as battery will be placed under the ground, moisture and cold will drop it's performance. Another issue is that for protecting battery health, it's charge should be kept around %80 percent. To ensure that a solar panel and a simple charging circuit is planned to be used. By these cautions, power will not be a problem for the system.

Instructions:



Motivation / Societal Impact (200–500 words)

Instructions: In 200–500 words, explain why this problem matters for the greater good (societal impact, safety, sustainability, accessibility, cost reduction, reliability, etc.). Include who benefits and how you would measure impact.

This project is important for several reasons. One of them is water is generally given for limited times of the days at the hobby gardens. Due to that reason, people have to be present in their garden at those times in order to open/close the valves. This is problematic for couple of reasons. Hobby gardens generally located in distance places, especially in the times like spring/summer where daily watering of plants is important, it is not feasible for people to do this task daily. Also generally people with daily jobs prefer to own hobby gardens, so they don't have time for daily visits. Another important problem is that as there are some offline working solutions, they generally lack robustness for situations such as pipes getting clogged or valves are not working properly and user can not be informed. I believe that the proposed solution will be a good choice for the problem mentioned. In top of all that, the solution is very cost friendly and easy to implement. In perspective of project's relevance to course, the components which will be used for the project are diversified. Motor drivers, communication modules, microcontrollers and sensors will be used and different interfaces will be implemented for them. By that, many topics will be covered in context of the class. Another good point is that as power consumption is crucial, optimization will be an important part of the project, as it is of the class.

Related Work / References

Add a few related works from the literature, patents, or existing products to show what other researchers and developers have done in this area (3–5 works are sufficient, but you may include more). At this stage, you are not expected to conduct a full literature review; that will be done after your project idea is accepted. ***Add entries in BibTeX format to `references.bib` and cite them in your text using `\cite{<bibtexhandle>}`****. The References will be ordered and generated automatically. You don't need to order them inside `references.bib` or delete references that you do not cite.*

Sample citation: Real-Time Operating Systems for Embedded Systems: A Survey [**?**].

References

[1] K. S. Enock, M. J. Sagali, U. I. Jeannick, and D. Chen, "Lora-based smart agriculture monitoring and automatic irrigation system," *Journal of Computer and Communications*, vol. 13, no. 3, pp. 1–20, 2025.

[2] S. Heble, A. Kumar, K. V. D. Prasad, S. Samirana, P. Rajalakshmi, and U. B. Desai, "A low power iot network for smart agriculture," in *2018 IEEE 4th World Forum on Internet of Things (WF-IoT)*, pp. 609–614, IEEE, 2018.

[3] S. S. Islam, M. M. Hasan, M. R. Islam, and M. F. Hossain, "Performance analysis of iot based solar operated smart water management system for irrigation field," in *2021 3rd International Conference on Electrical & Electronic Engineering (ICEEE)*, pp. 1–4, IEEE, 2021.

[4] X. Yu, P. Wu, W. Han, and Z. Zhang, "The research of an advanced wireless sensor networks for agriculture," *African Journal of Agricultural Research*, vol. 7, no. 5, pp. 851–858, 2012.

[5] S. Balivada, G. Grant, X. Zhang, M. Ghosh, S. Guha, and R. Matamala, "A wireless underground sensor network field pilot for agriculture and ecology: Soil moisture mapping using signal attenuation," *Sensors*, vol. 22, no. 10, p. 3913, 2022.

How to Submit

- Open the Google Drive sheet (submission tracker): `https://docs.google.com/spreadsheets/d/17ACM2EVBZ9tZY3XIKveWwRSbt-QfVjS7qkYcTWC3GwU/edit?usp=sharing`
- Fill in your group and project details in an appropriate row (make sure you do not delete/edit other students' entries!).
- Create a repository that you will use to submit **all** project work throughout the semester (GitHub or GitLab).
- Upload/commit your proposal files to the repository (both the `.zip` source files with `.tex` and the compiled `.pdf`).
- Paste the repository link into the corresponding column in the sheet.
- If your repository is private, add the instructor as a collaborator: `selmadilek@gmail.com`.